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Traffic Accident Analysis

USING POWER BI

A Business Intelligence Project Report

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Abstract

This project presents a comprehensive analysis of road accident data using Microsoft Power BI. The objective is to understand accident patterns, casualty trends, and risk factors through interactive data visualization. The dashboard examines Current Year (CY 2022) versus Previous Year (PY 2021) casualty trends, vehicle-wise breakdown, road type distribution, weather and light condition impacts, and geographic spread across Indian states. With 1,250 records spanning 2020–2023, the analysis reveals that total CY casualties stood at 1,550 — a marginal reduction of 0.7% from PY — while accidents increased by 7.9%, highlighting the need for targeted road safety interventions.

Introduction

Road accidents are a leading cause of preventable deaths globally, generating enormous volumes of incident data that, if properly analyzed, can drive evidence-based safety policies. Business intelligence tools like Microsoft Power BI enable the transformation of raw accident records into interactive dashboards accessible to traffic authorities, policy makers, and researchers without requiring deep technical expertise.

This project analyzes an Indian road accident dataset covering multiple states, vehicle types, road categories, weather conditions, and time periods. The goal is to identify high-risk scenarios — whether by vehicle type, road classification, time of day, or geography — and provide clear, actionable visual insights to support targeted safety improvements.

Problem Statement

Despite the availability of detailed accident records, traffic authorities often lack structured tools to extract patterns from raw data. Key questions that remain unanswered include:

- Which vehicle types and road categories are responsible for the most casualties?
- How do weather and time-of-day conditions influence accident severity?
- Which states and regions show the highest concentration of accidents?
- How have casualty numbers trended month-over-month across current and previous years?
- What proportion of accidents result in high vs. medium vs. low severity outcomes?

This project addresses these gaps by building an interactive Power BI dashboard that maps raw accident records into clear visual patterns, enabling data-driven road safety decision-making.

Objectives

- Analyze CY 2022 vs PY 2021 casualty and accident counts with percentage change
- Visualize monthly casualty trends to identify seasonal risk peaks
- Break down casualties by vehicle type, road type, and accident severity
- Assess the influence of weather conditions and time of day on casualties
- Identify high-risk states and geographic hotspots using state-level data
- Provide interactive filtering by Road Surface and Weather Conditions
- Build a multi-visual dashboard for traffic management and policy stakeholders

Dataset Description

The dataset contains 1,250 road accident records spanning 2020–2023 across multiple Indian states. Each record captures a unique accident event with the following key attributes:

Attribute	Description
Accident_ID / Date	Unique accident identifier and date of occurrence
Year / Month / Day_of_Week	Temporal dimensions for time-series and day-pattern analysis
State / City	Geographic location of the accident
Accident_Severity	Severity classification: High, Medium, Low
Number_of_Casualties	Total casualties per accident event
Weather_Condition	Prevailing weather: Clear, Rainy, Foggy, Hazy, Cloudy, Stormy
Road_Type	Road classification: National Highway, State Highway, Expressway, Urban, Rural, Service
Vehicle_Type	Vehicle involved: Car, Two-Wheeler, SUV, Truck, Bus, Tempo, Auto-Rickshaw
Time_of_Day	Time band: Morning, Afternoon, Evening, Night, Late Night
Is_Fatal	Binary flag indicating whether the accident was fatal (1) or not (0)

Tools and Technologies

Microsoft Power BI Desktop

Primary tool used for data import, Power Query transformations, DAX measure creation, and building the interactive Road Accident Analysis dashboard with slicers for Road Surface and Weather Conditions.

DAX (Data Analysis Expressions)

Used to compute: Total CY/PY Casualties, Total CY/PY Accidents, High/Medium/Low Severity counts, year-over-year percentage changes, and month-level aggregations for trend analysis.

CSV Data Source

The accident dataset was provided in CSV format and imported via Power BI's "Get Data" connector. Power Query Editor was used for cleaning and deriving calculated columns.

Methodology

Data Import — Road_Accident_Data.csv loaded into Power BI Desktop via "Get Data". Power Query Editor used for initial schema inspection and type validation.

Data Cleaning — Verified column data types, handled null values, standardized categorical labels (Severity, Vehicle Type, Road Type). Confirmed date parsing for Year, Month_Num fields.

DAX Measures — Created Total CY Casualties (1,550), Total PY Casualties (1,561), CY Accidents (327), PY Accidents (303), High/Medium/Low severity breakdowns, and YoY % change measures.

Visualizations — Built KPI cards (5 metrics), area/line chart (monthly trend), bar charts (vehicle type, road type), donut charts (severity, time of day), and geographic state-level visuals.

Dashboard Design — Single-page dark-themed dashboard with Road Surface and Weather Condition slicers. Color-coded KPI cards with YoY % indicators. All visuals linked to slicers for dynamic filtering.

Dashboard Analysis

Key Performance Indicators (CY 2022)

1,550	327	99	120	108
Total CY Casualties	Total CY Accidents	High Severity	Medium Severity	Low Severity

KPI Metric	CY 2022	PY 2021	YoY Change
Total Casualties	1,550	1,561	-0.7%
Total Accidents	327	303	+7.9%
High Severity Casualties	977	1,013	-3.6%
Medium Severity Casualties	476	458	+3.9%
Low Severity Casualties	97	90	+7.8%
Fatal Accidents (Is_Fatal=1)	99	98	+1.0%

Monthly Casualty Trend (CY 2022 vs PY 2021)

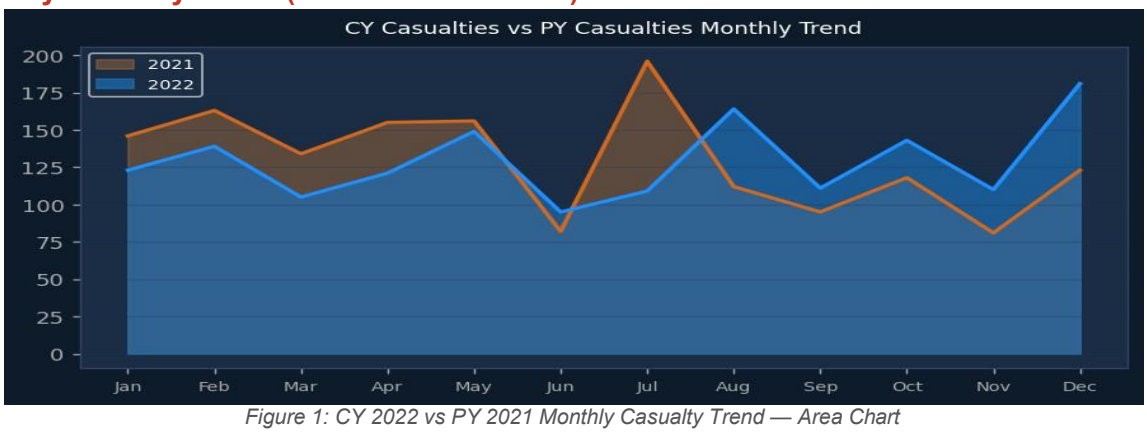
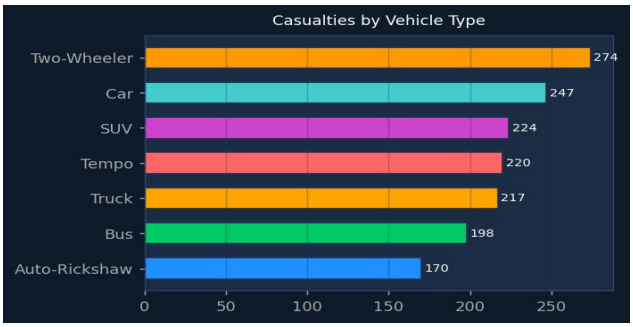


Figure 1: CY 2022 vs PY 2021 Monthly Casualty Trend — Area Chart

December (181) and August (164) were the peak casualty months in CY 2022, while July was the PY peak (196). The CY trend shows higher casualties in H2 (Jul–Dec: 918) versus H1 (Jan–Jun: 632), suggesting seasonal risk escalation in the second half of the year.



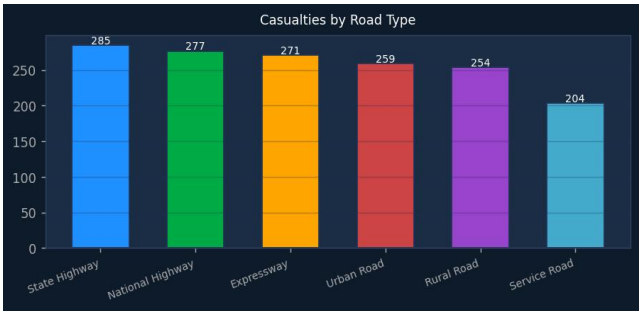


Figure 2: Casualties by Vehicle Type (left) | Casualties by Road Type (right)

Vehicle Type: Two-Wheelers (274) and Cars (247) are the most casualty-prone vehicle categories, together accounting for 34% of CY casualties. SUVs (224) and Tempos (220) closely follow. **Road Type:** State Highways (285) record the highest casualties, followed by National Highways (277) and Expressways (271) — high-speed corridors collectively contributing 54% of total casualties.

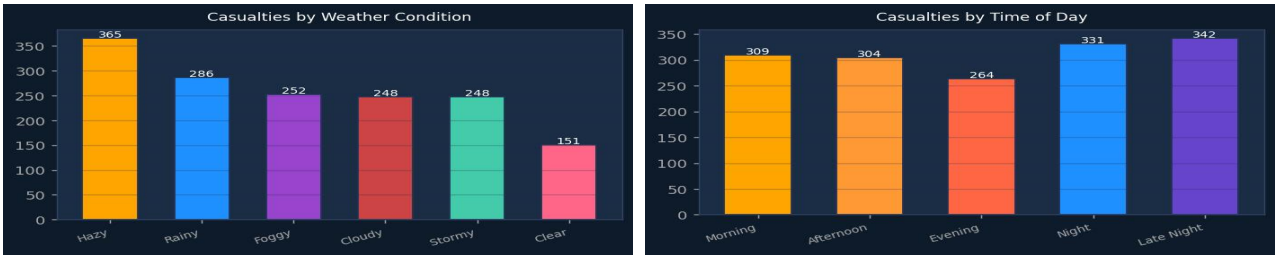


Figure 3: Casualties by Weather Condition (left) | Casualties by Time of Day (right)

Weather: Hazy (365), Rainy (286), and Foggy (252) together cause 58% of CY casualties — poor-visibility is the primary environmental risk. **Time of Day:** Late Night (342) and Night (331) are deadliest, accounting for 43% of casualties, driven by low visibility and driver fatigue.

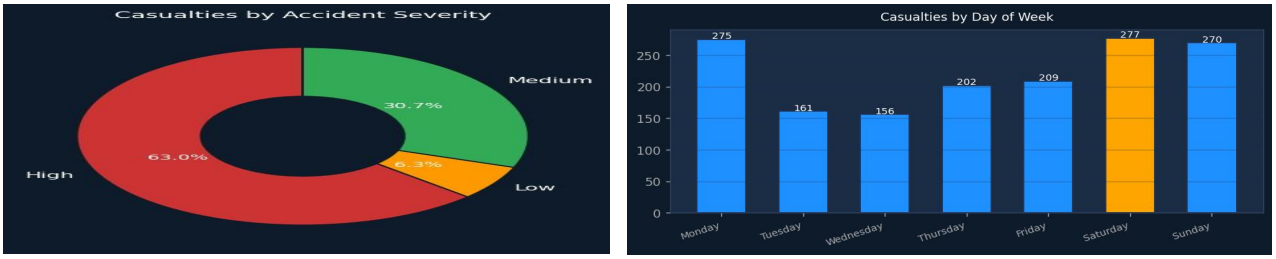


Figure 4: Casualties by Accident Severity (left) | Casualties by Day of Week (right)

Severity: High-severity accidents form 63% of CY casualties (977/1,550). **Day of Week:** Saturday (277), Monday (275), and Sunday (270) peak — weekend leisure travel and fatigue are primary drivers.

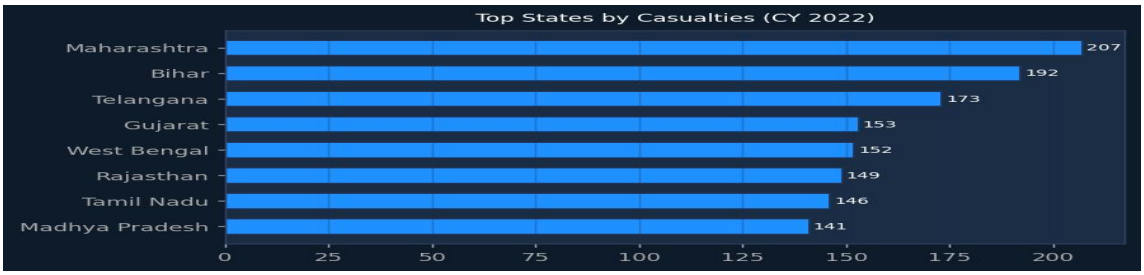


Figure 5: Top States by Casualties (CY 2022)

Maharashtra (207), Bihar (192), and Telangana (173) lead CY casualties, collectively accounting for 37% of national total — geographically concentrated risk requiring targeted interventions.

Key Insights and Findings

- CY casualties (1,550) fell 0.7% vs PY but accidents rose 7.9% — more frequent, slightly less lethal incidents.
- Two-Wheelers (274) and Cars (247) are the most vulnerable vehicle categories — together 34% of casualties.
- State Highways (285) and National Highways (277) are the deadliest road types — 54% of CY casualties.
- Hazy (365), Rainy (286), Foggy (252) poor-visibility conditions cause 58% of all CY casualties.
- High-severity accidents form 63% of CY casualties; Late Night and Night hours contribute 43% of total.
- Maharashtra (207), Bihar (192), Telangana (173) are top-3 states — 37% of national CY casualties.

Conclusion

This project demonstrates how Power BI transforms raw road accident records into actionable intelligence. The dashboard provides macro-level KPI tracking and micro-level pattern analysis across vehicle types, road categories, weather, time of day, and geography. Key findings — poor-visibility conditions, nighttime driving, and high-speed road corridors as primary risk factors, with Maharashtra, Bihar, and Telangana as hotspot states — can directly inform traffic policy, infrastructure investment, and public safety campaigns.

References

- Microsoft Power BI Documentation — https://docs.microsoft.com/en-us/power-bi/
- DAX Reference Guide — Microsoft Learn
- Road Accident Dataset — Kaggle / Government Open Data
- Ministry of Road Transport & Highways — Annual Road Accident Report